METHODOLOGY REPORT

# Congestion and Bottleneck Identification (CBI)

## Arterial Analysis: Methods, Data and Results

**How the arterial study works, how it developed, and how far its results can be trusted**

| | |
|---|---|
| **Organization** | Atlanta Regional Commission (ARC) |
| **Edition** | November 2026 |
| **Study year** | 2025 probe-speed data |
| **Scope** | Arterial roads in the ARC modeling region |

This report explains the arterial delay analysis (script `cbi_arterial_intersection_severity`), the data it uses, the decisions made along the way and the checks performed. It is written for planners and decision makers; technical detail is kept in formulas, tables and the appendices.

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## Executive summary

The Atlanta Regional Commission (ARC) measures where and when arterial roads in the region are congested, how much delay that congestion causes, and what kind of daily pattern it follows. This report explains how that analysis works and how it reached its current form.

**In plain terms**

We take a year of vehicle speeds measured every five minutes on every arterial road segment, compare each segment's speed with how fast it moves when the road is empty, and turn the slowdown into hours lost by drivers, using estimates of how many vehicles use the road at each hour. We then rank road sections by how much delay they cause per mile, find the worst one-mile stretches, and group roads by the shape of their typical day.

**What changed.** The study began with a federal speed dataset (NPMRDS) that covers only the National Highway System, and with a road list in which many arterials had no classification. Both limits hid real arterials. The current analysis uses a full-network speed product for 2025 and a road list completed with three independent checks.

| | **Before expansion** | **Current run** |
|---|---|---|
| Arterial road segments (TMCs) | about 4,550 | about 8,690 |
| Directional arterial miles | 3,962 | about 6,415 |
| Speed records used | NPMRDS (NHS only) | 2,028,300,831 (full network) |
| Days of the week | weekdays, then all days | all 365 days, weekday and weekend reported separately |
| Congestion-pattern groups | k-means, separation 0.297 | hierarchical, separation 0.339 |

**Headline results (2025).** Across about 6,415 directional miles of arterials:

| | |
|---|---|
| Estimated delay, average weekday | **395,804** vehicle-hours per day |
| Estimated delay, average day (all seven days) | **359,389** vehicle-hours per day |
| Estimated delay, average weekend day (derived) | about **268,000** vehicle-hours per day |
| Estimated annual delay | **131.2 million** vehicle-hours |

Half of the arterial mileage is mostly uncongested; congestion is concentrated on a smaller share of roads, much of it in the evening peak (Section 9).

**What the results are.** Speeds are measured (by the data vendor). Traffic volumes and delay are *estimated*: they combine modeled daily traffic with a regional hour-by-hour profile. The results are therefore a consistent way to compare and prioritize roads, not a direct count of the hours drivers lost.

**What is still needed.** The main open task is an independent check of the estimated hourly traffic against count stations that were not used to build the estimates, followed by field confirmation of the top-ranked locations (Section 14).

# 1 How the study developed

The arterial analysis grew in five phases. Each one fixed a limit that the previous phase exposed.

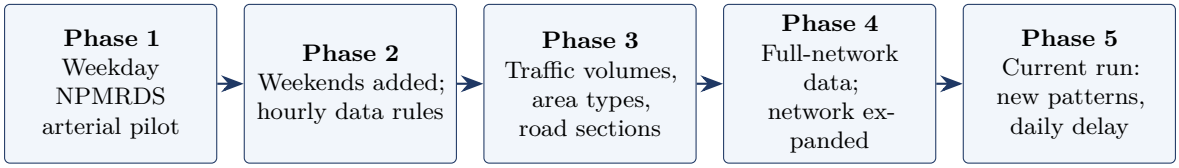



**Figure 1:** The five development phases of the arterial analysis.

## 1.1 Phase 1: a weekday pilot on NPMRDS data

The first version used NPMRDS, the federal probe-speed dataset, on weekdays only. It set up the core steps that are still used: pick the arterial segments (functional classes 3 and 4, principal and minor arterials), group them by road and direction, compare each segment's speed with its own free-flow speed, convert the slowdown into delay, and rank road sections.

Two limits sat *before* the analysis, not inside it:

- **Coverage.** NPMRDS describes the National Highway System. Arterials outside it had little or no data.
- **Classification.** A segment with no functional class in the road database never entered the arterial list, even when it was plainly an arterial (Peachtree Street downtown was the first case found).

The pre-expansion network had about 4,550 segments and 3,962 directional miles. Only about half a mile of *that* list was left out of the analysis, but that says nothing about arterials missing from the list itself.

## 1.2 Phase 2: weekends and hourly data-quality rules

The analysis was extended to all seven days, with weekday and weekend results reported separately. At first this exposed a gap: 31% of arterial-corridor segments had no weekend data until a weekend supplement was loaded.

Weekend and night-time data also revealed a problem on very short segments. On a 200-foot segment at a traffic signal, an hour with only one or two speed readings often records a single car waiting at a red light. That made empty roads look congested at 2 AM (Section 7.2). Two rules now prevent this:

- an hour of one day counts only if it has at least **4 of its 12** five-minute readings;
- a "typical" speed for an hour pools every reading for that hour across the year (at least 30 readings).

### 1.3 Phase 3: traffic volumes, land-use context and road sections

Delay depends on how many vehicles are slowed, so the study added traffic volumes: Replica's estimated directional daily traffic as the main source, a regional hour-by-hour traffic profile from Georgia's continuous count stations, and NPMRDS/HPMS traffic as a fallback. Each segment also received its ARC Activity-Based Model (ABM) area type, and roads were cut into sections where the street name or area type changes. A rolling one-mile "hotspot" measure was added beside the section ranking.

### 1.4 Phase 4: full-network data and a complete arterial list

A full-network speed product obtained from Probe Data Analytics of RITIS for 2025 (a 19-county and a 2-county download) replaced NPMRDS for the arterial analysis: 2,028,300,831 five-minute readings on 19,834 segments. A three-way check (HERE road class, OpenStreetMap road class and traffic volume) then added 2,297 previously unclassified segments (migration 021), and a follow-up (021b) kept their short connecting pieces while excluding pieces that failed the check.

### 1.5 Phase 5: the current run

The analysis was rerun on the complete network and the new data. Delay is now annualized from an average day, average weekday and average daily delay are reported, area types use the correct ABM definition, and the four congestion patterns are now formed by hierarchical clustering, which separated them slightly better than k-means on this data.

**Table 1:** What each phase added, and what to keep in mind when comparing results across phases.

| Phase | Main capability | Caution when comparing |
|---|---|---|
| 1 Weekday pilot | NPMRDS, restricted arterial list, weekday profiles | Limited coverage and days |
| 2 Weekends | Weekend results; minimum-readings and pooled-speed rules | Weekend data initially sparse |
| 3 Volumes and sections | Replica traffic, hourly profile, ABM area types, sections, hotspots | Delay depends on estimated traffic |
| 4 Full network | 2.03 billion readings; 021/021b network expansion | New speed product is not interchangeable with NPMRDS |
| 5 Current run | Annualized delay, daily delay, hierarchical patterns | Different data, network and features from earlier runs |

**Read with care**

Results from different phases are **not directly comparable**: the road list, the speed source and some calculations changed. Compare rankings only on the roads common to both runs (Section 8).

# 2 Terms used in this report

| Term | Meaning |
|---|---|
| TMC | Traffic Message Channel segment: a short, one-direction piece of road (typically 0.1–2 miles) to which speeds are reported. It is not a signal or a count station. |
| NPMRDS | National Performance Management Research Data Set: federal probe-speed data, mainly for the National Highway System (NHS). |
| Full-network product | The 2025 probe-speed download covering every TMC, including all arterials, used for the arterial analysis. |
| Free-flow speed | How fast a segment moves when traffic is light, measured from its own speeds at the region's fastest hour (4 AM). |
| Speed ratio | Speed divided by free-flow speed. 1.0 = free flow; below 0.70 = congested. |
| AADT | Average Annual Daily Traffic: vehicles per day, averaged over the year; "directional" means one direction only. |
| Replica | A commercial travel model that estimates traffic on every road; source of directional AADT. |
| HDF | Hourly Distribution Factor: the share of a day's traffic in each hour, from Georgia's continuous count stations (TMAS). |
| Excess travel time | Extra time a vehicle spends crossing a segment because it is slower than free flow. |
| EAVHD | Estimated Annual Vehicle-Hours of Delay: excess travel time × estimated vehicles, summed over the year. |
| Recurrence | Share of days on which a segment is congested during the AM (7–9) or PM (4–6) peak. |
| Section | A stretch of one named road with one area type, at least 1 mile long, both directions combined. |
| Hotspot | The worst one-mile window of delay along a road. |
| COP | Congestion Operating Pattern: one of four groups of segments with a similar daily speed shape. |
| ABM area type | ARC Activity-Based Model land-use class of the zone (TAZ) a segment lies in: CBD to Rural. |
| FRC | HERE Functional Road Class (1 = highest). |
| OSM | OpenStreetMap road class (trunk, primary, secondary, . . . ) as carried in Replica. |
| FDR | False Discovery Rate: a correction for running many statistical tests at once. |

# 3 Data sources

**In plain terms**

Speeds come from one source for each analysis and are never mixed. Traffic volumes are estimates. Area types come from ARC's travel model. Each input does one job.

**Table 2:** Inputs, their role, and what they do and do not measure.

| Input | Role | What it is |
|---|---|---|
| Full-network probe product (2025) | Arterial speeds, every 5 minutes, every TMC | Vendor speeds; sparse intervals filled and smoothed |
| NPMRDS (2025) | Freeways, corridor bottleneck pipeline and its overnight test | Probe speeds, NHS only |
| Replica directional AADT | Main traffic volume for each arterial TMC | Modeled daily traffic |
| NPMRDS/HPMS AADT | Fallback volume where no good Replica match exists | Two-way estimate; halved per direction |
| GDOT TMAS counts | Regional hour-by-hour traffic profile (HDF) | Counted at continuous stations |
| ARC ABM zones (TAZ) | Area type of each TMC | Model land-use class |
| HERE network + Replica/OSM | Road classification checks and road shapes | Road attributes |

**One source per analysis.** The arterial analysis and the Arterial Speed Explorer read the full-network product. The corridor bottleneck pipeline, its overnight eligibility test and the Freeway Speed Explorer keep NPMRDS. The full-network table holds TMC, time, speed, travel time, the vendor's reference speed and a confidence score (0.70–0.99). It has no historical-average speed or data-density field; nothing in the arterial analysis uses them.

## 3.1 How the full-network product compares with NPMRDS

Where both products have a reading for the same TMC and the same five minutes, they were compared directly.

| Check | Result | What it means |
|---|---|---|
| Speed agreement | Average difference 7.1 mph; 51% of pairs differ by more than 5 mph, at every confidence level | Different products: NPMRDS averages the few vehicles present; the full-network product fills and smooths |
| Time alignment | Smallest difference with no time shift (Figure 2) | Timestamps agree; no correction needed |
| Reference speed | Equal on only 1.2% of readings; full-network reference about 8.0 mph lower | "Free flow" is defined differently, which is why V3 measures its own |
| Confidence score | Agreement similar from 0.70 to 0.99 | Not used as a filter; all readings kept |

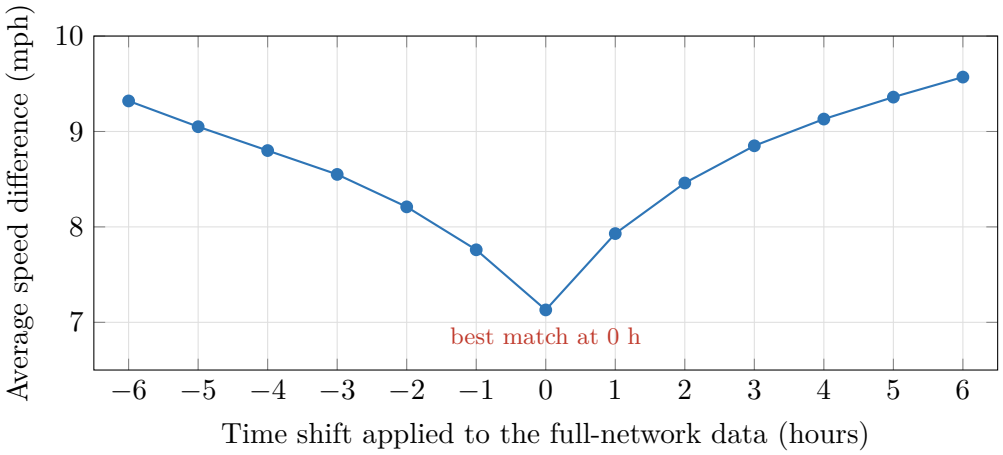



**Figure 2:** Matching the two products after shifting one in time. The disagreement is smallest with no shift, so the timestamps line up; the remaining 7 mph difference comes from the products themselves.

**Read with care**

Neither product is "ground truth". Filling and smoothing in the full-network product may soften brief, sharp queues. NPMRDS, on the other hand, can over-react when only one or two vehicles are observed.

## 3.2 Completeness of the full-network data

The permanent table holds **2,028,300,831** readings on **19,834** TMCs, from 1 January 2025 00:00 to 31 December 2025 23:55. The theoretical maximum is 19,834 TMCs × 288 five-minute intervals = 5.71 million readings a day.

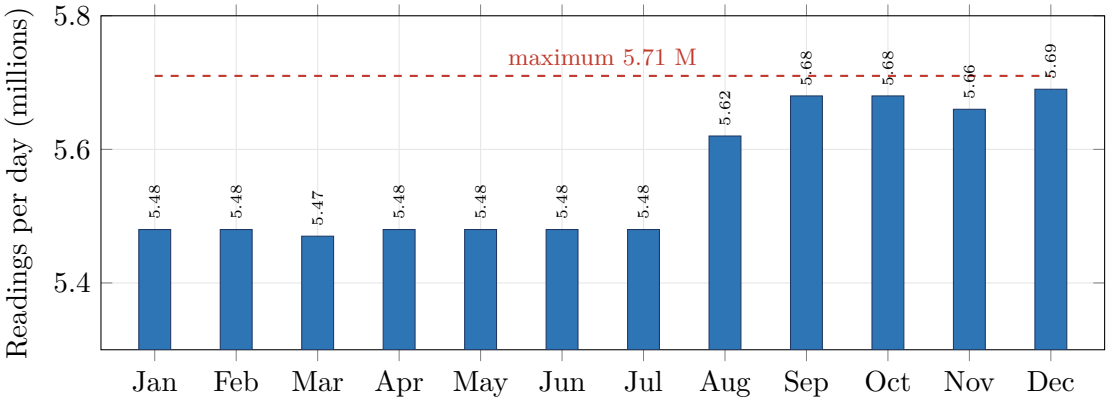



**Figure 3:** Readings per day by month: about 96% complete from January to July, 98–99.6% from August. The step in August most likely reflects TMCs that became active part-way through 2025.

Because some TMCs have fewer months of data, delay is annualized from the *average day* rather than summed (Section 7.4); otherwise a TMC with five months of data would show about 58% too little annual delay.

**Table 3:** Readings loaded per month.

| Month | Readings | Month | Readings | Month | Readings |
|---|---|---|---|---|---|
| Jan | 169,893,097 | May | 169,929,327 | Sep | 170,368,843 |
| Feb | 153,373,432 | Jun | 164,435,987 | Oct | 176,015,288 |
| Mar | 169,614,650 | Jul | 169,959,355 | Nov | 169,675,069 |
| Apr | 164,390,183 | Aug | 174,256,018 | Dec | 176,389,582 |
| **Total** | | | | | **2,028,300,831** |

# 4 Which roads count as arterials

**In plain terms**

A road segment is analysed as an arterial when its functional class is 3 or 4 (principal or minor arterial). Many segments had no class at all, so real arterials were missing. We checked those segments against three independent signals and added the ones that passed all three.

## 4.1 How a segment gets its class

Each TMC's class is decided in this order:

1. an **override** recorded by ARC (table `tmc_class_override`), with the reason stored;
2. otherwise the TMC's **own class** from the NPMRDS road database;
3. otherwise a class **borrowed** from another TMC on the same route (`tmclinear`); when several are available, the highest class number (lowest road class) is taken.

Interstate-named segments are never arterials.

## 4.2 The gap: unclassified state routes and streets

A network-wide audit sorted every TMC into a category. The arterial list itself was analysed almost completely (3,962 miles, only 0.5 mile left out), but about **1,700 miles of US and state routes** and **4,600 miles of named streets** had no class anywhere, so they could never enter the analysis.

## 4.3 Three independent checks

Two road classifications were compared inside ARC's database, using each TMC's actual road shape: **HERE Functional Road Class (FRC)** from the licensed TMC network file, and the **OpenStreetMap class** carried by Replica's road network.

**Matching method.** Three points are placed along the TMC (at 25%, 50% and 75% of its length). At each point the nearest Replica road within 20 m, travelling the same way (heading within 45°), is found. The TMC "lies on" an OSM arterial road (trunk, primary or secondary) if at least two of the three points agree.

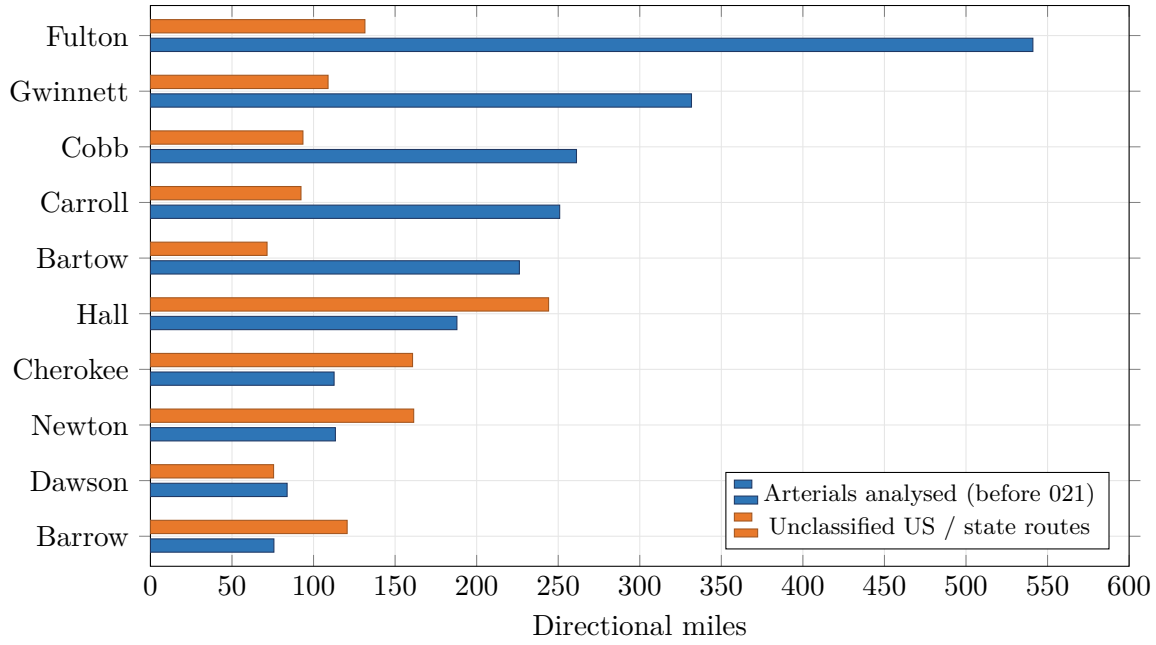



**Figure 4:** Arterial miles analysed before the expansion and unclassified US/state-route miles, selected counties. In several exurban counties (Hall, Newton, Cherokee, Barrow) more state-route mileage was unclassified than analysed.

**Is the matching reliable?** It was tested on the 3,898 miles of arterials already known. Using straight lines between each TMC's end points failed on long, curving segments; using the real road shapes matched 91–99% in every length class (Figure 5).

**Why one classification alone is not enough.**

- 41% of known arterial miles are FRC 4, so "FRC 2–3 only" would drop four in ten real arterials.
- Within FRC 4, known arterials lie on OSM arterial roads 94% of the time, so the OSM check separates arterials from collectors.
- HERE gives freeway ramps the class of the road they serve; ramps almost never match OSM arterial roads, so the OSM check also keeps ramps out.

**The third check: traffic volume.** Replica's directional traffic on the matched road was compared with the known arterials. A floor of 1,500 vehicles a day per direction removes only 2.6% of known arterial miles; higher floors start removing real (mostly rural) arterials.

## 4.4 The rule (migration 021) and its result

A TMC is added as an arterial only when **all four** hold:

1. it has no class today (no override, no own class, no same-route class), has a road name, is not Interstate-named, and is at least 0.1 mile long;
2. its HERE FRC is 2, 3 or 4;

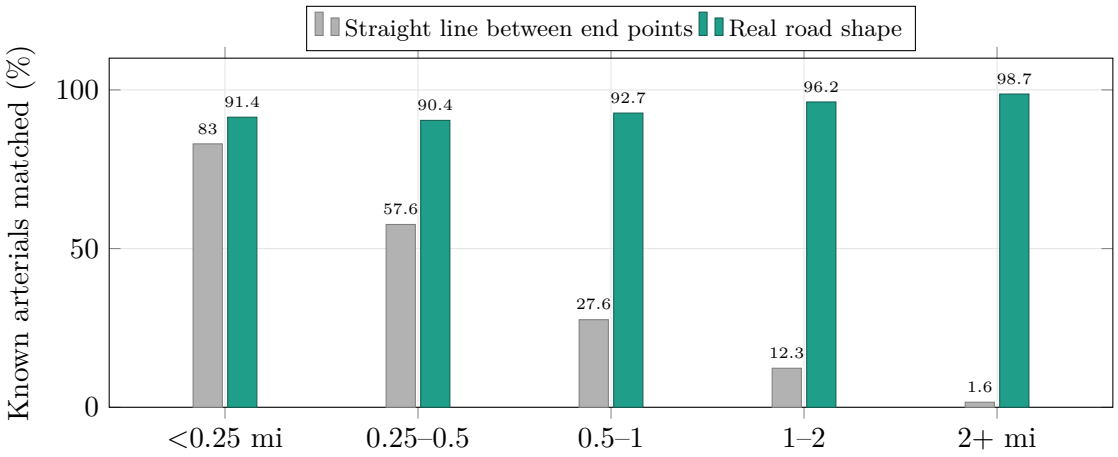



**Figure 5:** Matching benchmark on known arterials, by TMC length. With real road shapes about 97% of known arterial miles are matched overall.

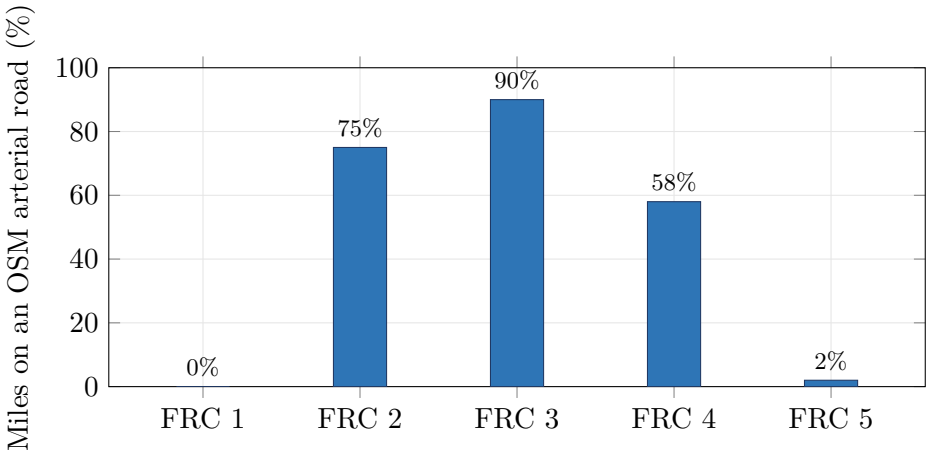



**Figure 6:** Share of each HERE road class lying on an OSM arterial road. FRC 4 is mixed (58%): it contains both minor arterials and collectors. Miles: FRC 1 = 42, FRC 2 = 703, FRC 3 = 2,478, FRC 4 = 6,754, FRC 5 = 170.

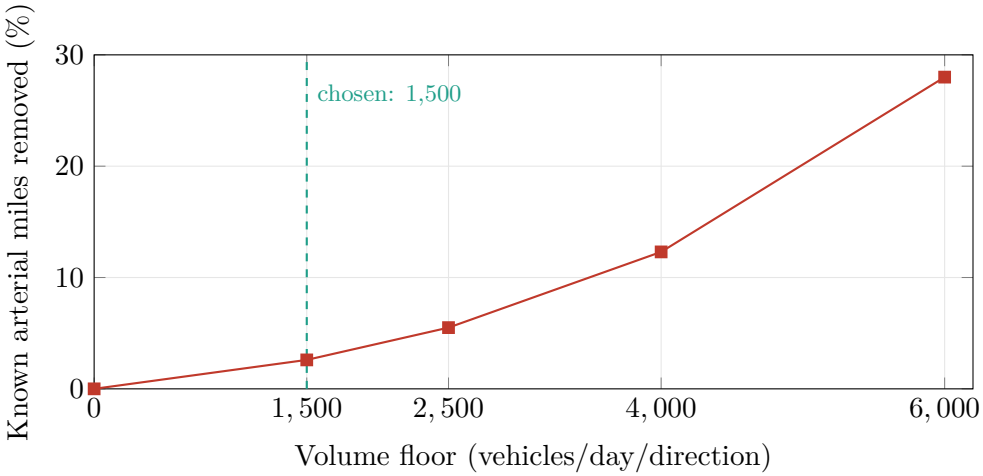



**Figure 7:** How much of the known arterial network each volume floor would wrongly remove.

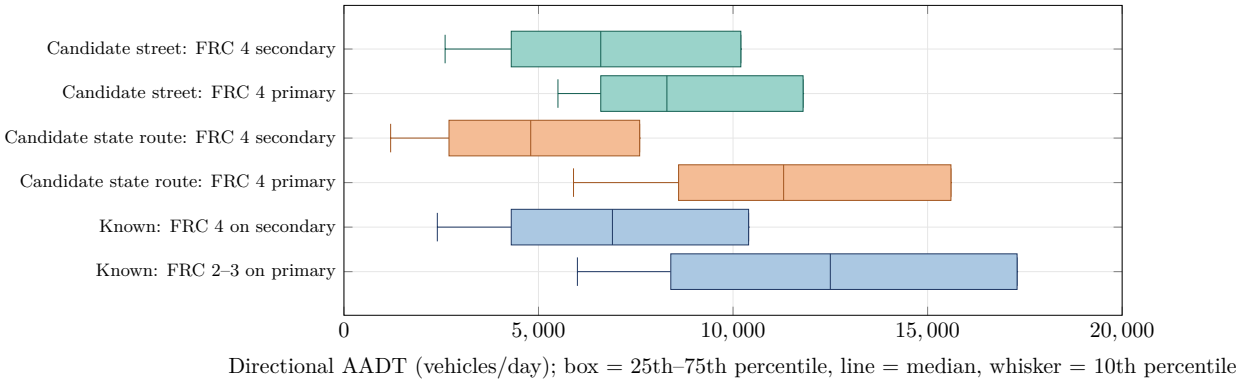



**Figure 8:** Traffic ranges of known arterials (blue) and candidates (orange: state routes; green: named streets). Named-street candidates look like known arterials. Rural state routes on OSM secondary roads run lower and are the least certain group.

3. it lies on an OSM trunk, primary or secondary road (not a ramp link), same direction, at 2 or more of 3 points;
4. its Replica directional traffic is at least 1,500 vehicles a day.

**Table 4:** Segments added by migration 021.

| HERE FRC | TMCs | Directional miles |
|---|---|---|
| FRC 2 | 16 | 18 |
| FRC 3 | 341 | 345 |
| FRC 4 | 1,940 | 2,024 |
| **Total** | **2,297** | **2,387** |

Every added segment received its Replica volume, street name and area type, and every override stores its evidence (for example "021: HERE FRC 4 + OSM secondary (3/3 pts) + Replica 6,200 veh/day/dir"). Two delete statements undo it.

## 4.5 Migration 021b: an unintended side effect, fixed

Because segments borrow the class of others on the same route, 021's additions also passed an arterial class to 2,318 other segments (617 miles) on those routes. Migration 021b restored the intended network:

- **1,847 short connectors** (under 0.1 mile, 66 miles) between the new arterial pieces stay arterial. Each receives an explicit override and the traffic, street name and area type of its nearest 021 neighbour.
- **471 longer segments** (551 miles) that had failed 021's own rule receive a collector override (class 5) and stay out.

The explicit override on connectors matters: because borrowing takes the *highest* class number on the route, a class-5 sibling would otherwise pull the connectors out again.

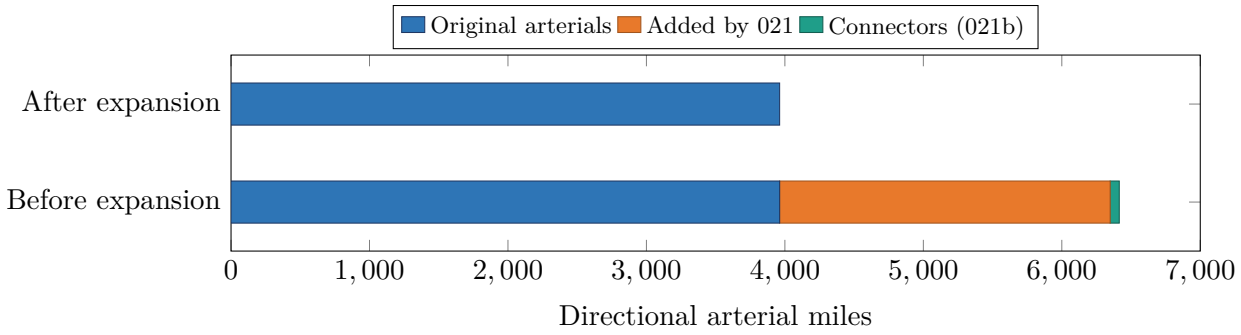



**Figure 9:** Growth of the arterial network: 3,962 to about 6,415 directional miles (about 4,550 to about 8,690 TMCs).

**Read with care**

Open classification items: about 170 miles carry an official *local* class but lie on OSM arterial roads with arterial-level traffic; 1,246 TMCs have no HERE FRC (mostly newer TMCs and ramp pieces); and a stricter 2,500 floor for rural state routes on OSM secondary roads remains an option to test. None of these were silently reclassified.

# 5 Area types and road sections

## 5.1 Area type: the corrected definition

Each TMC takes the area type of the ARC ABM zone (TAZ) it lies in. The ABM Data Dictionary (February 2022) defines area type twice: once for highway links (`ATYPE`, by density) and once for zones (`AREATYPE`, by land use). Because TMCs take their zone's value, the **zone definition** is the right one. Earlier editions used the link labels.

**Table 5:** ABM area types: current (zone) labels and earlier (link) labels.

| Code | Zone AREATYPE (used now) | Link ATYPE (earlier label) |
|---|---|---|
| 1 | CBD | Very high density urban / CBD |
| 2 | Urban Commercial | High density urban |
| 3 | Urban Residential | Medium density urban |
| 4 | Suburban Commercial | Low density urban |
| 5 | Suburban Residential | Suburban |
| 6 | Exurban | Exurban |
| 7 | Rural | Rural |

Codes, section boundaries and results are unchanged by the correction; only the descriptions change. Summary groups are CBD (1), Urban (2–3), Suburban (4–5) and Exurban/Rural (6–7). The codes are categories, not a measured scale.

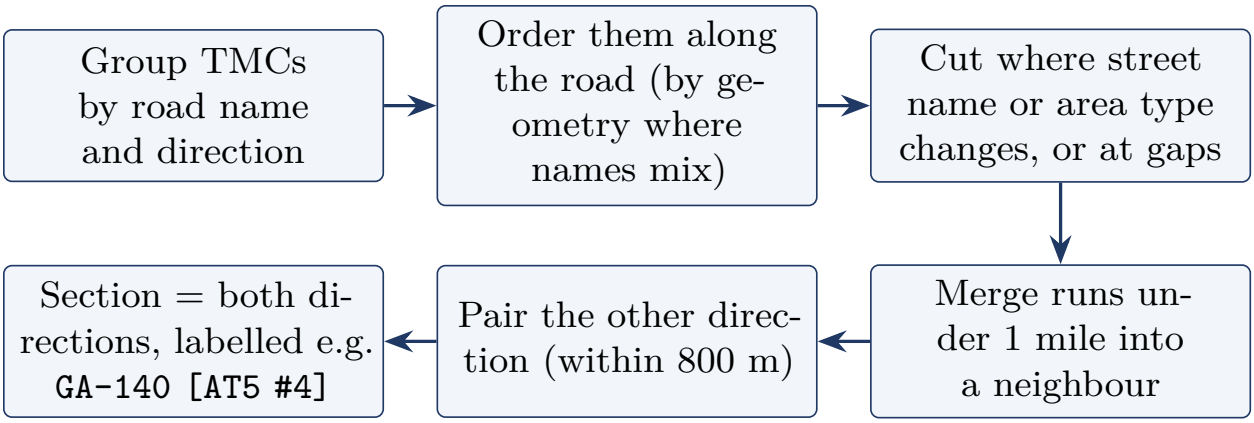



**Figure 10:** Section construction. Settings: split by area type and by street name, minimum section 1.0 mile, pairing distance 800 m.

### 5.2 How roads are cut into sections

Each TMC stays whole: one crossing a zone boundary is not divided. A section takes its mileage-dominant area type. Area type decides which segments are grouped; it does not change any speed, volume or delay.

## 6 Traffic volumes

**In plain terms**

Speed tells us how slow a road is; traffic volume tells us how many people that slowness affects. We take each segment's estimated daily traffic in its direction and spread it over the 24 hours with a regional hourly profile.

### 6.1 Daily traffic per direction

- **Main source: Replica 2025 directional AADT**, matched to each TMC by location and direction (table `replica_tmc_dir_aadt`). It is already one direction, so it is never halved again.
- **Fallback: NPMRDS/HPMS AADT** where no acceptable Replica match exists. This is a two-way figure, so each direction gets half:

$$\mathrm{AADT}_{i,\mathrm{dir}} = \mathrm{AADT}_{i,\text{2-way}} \times D, \qquad D = 0.50. \tag{1}$$

  The 50/50 split is an assumption, not a measured split.
- Segments added by 021 and 021b take their volume from the same Replica match that confirmed them as arterials. Every TMC records which source it used.

**Reading the table.** Replica supplies the traffic for 8,535 of 8,687 TMCs (98.3%) and 6,365 of 6,415 miles (99.2%). The NPMRDS fallback is used on only 152 TMCs (50 miles): 148 where

**Table 6:** Where each TMC's traffic volume comes from (current run: 8,687 TMCs, 6,415 directional miles).

| Volume source | TMCs | Miles | Share of miles |
|---|---|---|---|
| Replica 2025 directional (original arterials) | 4,125 | 3,793 | 59.1% |
| Replica 2025, 021 match: OSM secondary | 1,953 | 2,030 | 31.6% |
| Replica 2025, 021 match: OSM primary | 333 | 349 | 5.4% |
| Replica 2025, 021 match: OSM trunk | 11 | 8 | 0.1% |
| Replica 2025, full file: OSM tertiary, residential, unclassified | 221 | 94 | 1.5% |
| Replica 2025, adjacent 021 TMC (021b connectors) | 1,846 | 66 | 1.0% |
| Replica 2025, full file: OSM motorway, motorway link, trunk link | 46 | 25 | 0.4% |
| NPMRDS fallback: Replica more than 2× off (for review) | 148 | 50 | 0.8% |
| NPMRDS fallback: no Replica match | 4 | 0 | 0.0% |
| **Total** | **8,687** | **6,415** | **100%** |

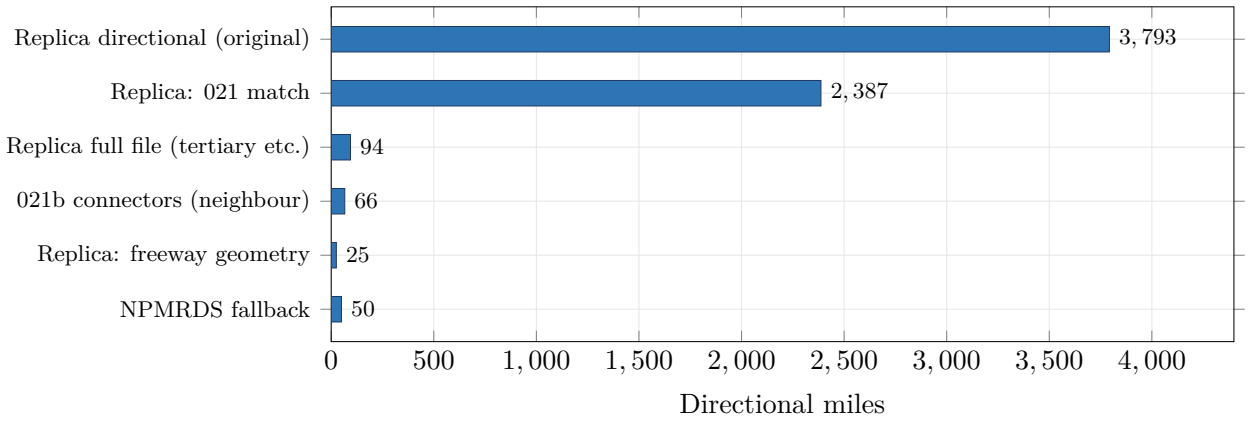



**Figure 11:** Volume sources by mileage. Replica supplies 99.2% of arterial miles; the NPMRDS fallback covers 0.8%.

Replica disagreed with the segment's own NPMRDS traffic by more than a factor of two (flagged for review) and 4 with no Replica match. A small group of 46 TMCs (25 miles) is classed as arterial by NPMRDS but lies on freeway geometry in OpenStreetMap (for example where a US route runs on a freeway); they are flagged and can be excluded from the arterial ranking if preferred.

### 6.2 Spreading daily traffic over the hours

The hourly distribution factor (HDF) gives the share of a day's traffic in each hour. It is built from GDOT's continuous count stations (TMAS). It uses the region's urban arterial stations, with faulty station-days removed and each station weighted so that stations with longer records do not dominate. For counted volume $V$ at station $s$, day $d$ and hour $h$:

$$\mathrm{HDF}_{s,d,h} = \frac{V_{s,d,h}}{\sum_{k=0}^{23} V_{s,d,k}}, \qquad \sum_{h=0}^{23} \mathrm{HDF}_h = 1. \tag{2}$$

The estimated traffic on TMC $i$ in hour $h$ is then

$$\widehat{V}_{i,h} = \mathrm{AADT}_{i,\mathrm{dir}} \times \mathrm{HDF}_h. \tag{3}$$

**Read with care**

The same regional profile is applied to every arterial and every day. Real roads differ (a mall road peaks at midday, a commuter road at 5 PM), and weekends differ from weekdays. Section 8 shows that the ranking barely changes under very different profiles, but the absolute delay totals depend directly on these volumes.

## 7 Free flow, congestion and delay

### 7.1 Free-flow speed

1. The region's **free-flow hour** is the hour with the highest average arterial speed: **4 AM**.
2. Each TMC's free-flow speed $S_{\mathrm{FF},i}$ is its own average speed in that hour, pooled over the year.
3. **Safety floor.** If the TMC's 85th-percentile hourly speed is more than 1.2 times that value (a sign of unrepresentative overnight readings), the 85th-percentile speed is used instead:

$$S_{\mathrm{FF},i} = \begin{cases} P85_i & \text{if } P85_i > 1.2\,\bar{S}_{i,\,4\mathrm{AM}} \\ \bar{S}_{i,\,4\mathrm{AM}} & \text{otherwise.} \end{cases} \tag{4}$$

The vendor's own reference speed is not used.

### 7.2 Hourly speed and the data-quality rules

For each TMC, day and hour, the analysis uses the **average of that hour's five-minute speeds**, but only if the hour has at least **4 of its 12** readings:

$$\bar{S}_{i,d,h} = \frac{1}{n_{i,d,h}} \sum_{t \in h} S_{i,t}, \qquad \text{used only if } n_{i,d,h} \geq 4. \tag{5}$$

**Why the rule exists: an example.** At Mableton Parkway and Veterans Memorial Highway, a 0.04-mile eastbound connector (TMC 101N12777) appeared congested on 27–41% of nights. Overnight it had only 1–2 readings an hour, and almost every "congested" night came from those thin hours: one car stopped at a red light sets the whole hour's average. The 0.74-mile westbound approach (101+12777), with more readings, shows the real pattern: little congestion at night and heavy congestion at 4–6 PM.

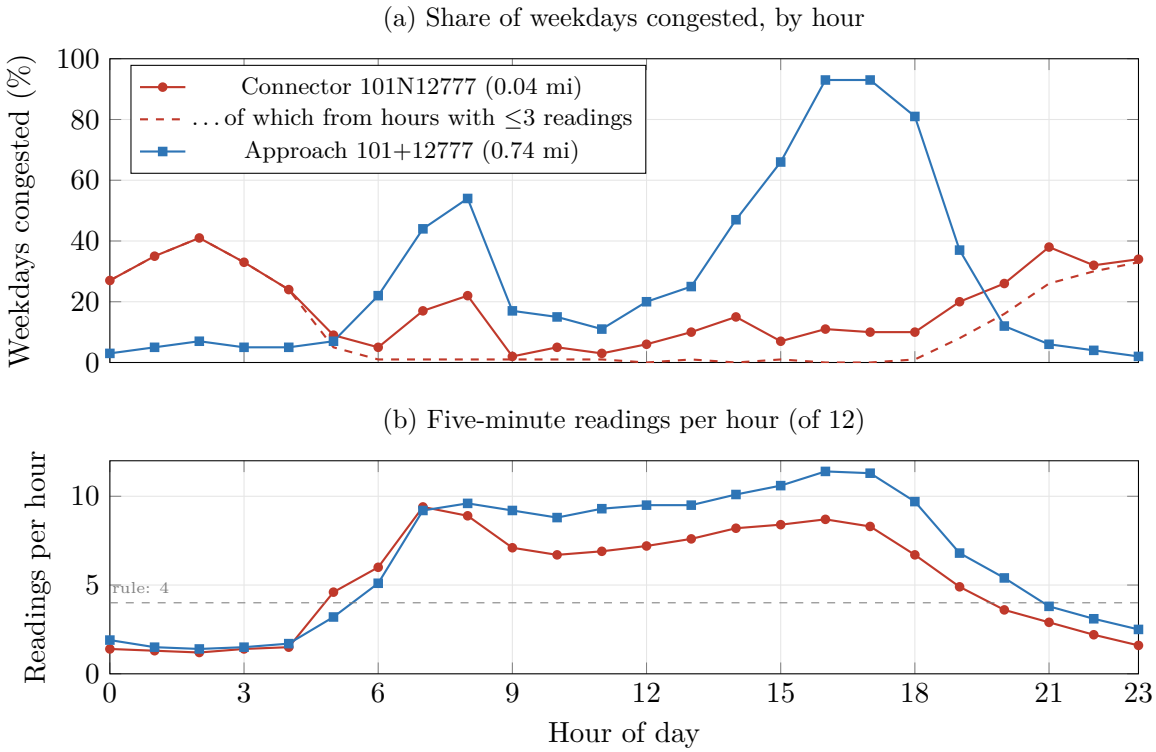



**Figure 12:** The sparse-hour problem on NPMRDS data at Mableton Pkwy & Veterans Memorial Hwy (weekdays). The connector's night "congestion" (solid red) is almost entirely made of hours with 3 or fewer readings (dashed red). The minimum-readings rule removes these hours.

On NPMRDS, the rule left out 130 TMCs (3.96 miles, median length about 120 feet), mostly intersection connectors with only 1.0–1.5 readings per peak hour. The full-network product has a reading in nearly every interval, so the rule now rarely applies, but it stays as a safeguard.

**Typical speeds for the maps** use a different, pooled rule: the median of every reading in that hour across all weekdays (or all weekend days), provided there are at least 30 readings. "Days congested" and daily delay are shown only where at least 5 days qualify.

## 7.3 Recurrence: how often a segment is congested

A day counts as congested when the slowest of its peak hours (7, 8, 9 AM or 4, 5, 6 PM) averages below 70% of free-flow speed:

$$\text{congested}_{i,d} = \Big[\min_{h\in\text{peak}} \bar{S}_{i,d,h} < 0.70\, S_{\text{FF},i}\Big], \qquad \text{Recurrence}_i = \frac{\#\text{congested days}}{\#\text{analysed days}}. \tag{6}$$

A TMC needs at least 30 analysed days to be included at all.

### 7.4 Excess travel time and annual delay

For each TMC, day and hour, the extra time per vehicle is

$$\mathrm{ETT}_{i,d,h} = L_i \times \max\left(0,\ \frac{1}{\bar{S}_{i,d,h}} - \frac{1}{S_{\mathrm{FF},i}}\right) \quad \text{(hours per vehicle)}, \tag{7}$$

where $L_i$ is the TMC length in miles. Any slowdown below free flow counts, not only slowdowns below 70%.

Annual delay is the **average day**, hour by hour, scaled to a full year, then multiplied by the traffic in that hour:

$$\mathrm{EAVHD}_i = \sum_{h=0}^{23} \Big(365 \times \underbrace{\tfrac{1}{N_{i,h}} \textstyle\sum_d \mathrm{ETT}_{i,d,h}}_{\text{average day}}\Big) \times \mathrm{AADT}_{i,\mathrm{dir}} \times \mathrm{HDF}_h, \tag{8}$$

where $N_{i,h}$ is the number of days with a valid hour $h$. Earlier runs *summed* over the days with data, which under-counted TMCs with fewer days (a TMC with five months of data would show about 58% too little delay).

#### Worked example (illustrative numbers)

A 0.5-mile segment with free-flow speed 40 mph averages 25 mph between 5 and 6 PM on a weekday: $\mathrm{ETT} = 0.5 \times (1/25 - 1/40) = 0.0075$ hours, or **27 seconds** per vehicle. If the segment carries 10,000 vehicles a day in that direction and 8% of them in that hour, 800 vehicles each lose 27 seconds: **6.0 vehicle-hours** of delay in that hour on that day.

### 7.5 Average weekday and average daily delay

Two figures in plain units (vehicle-hours per day) are reported for every TMC and section:

$$\text{Avg weekday delay}_i = \mathrm{AADT}_{i,\mathrm{dir}} \sum_h \mathrm{HDF}_h \times \overline{\mathrm{ETT}}^{\text{weekdays}}_{i,h}, \tag{9}$$

$$\text{Avg daily delay}_i = \mathrm{AADT}_{i,\mathrm{dir}} \sum_h \mathrm{HDF}_h \times \overline{\mathrm{ETT}}^{\text{all days}}_{i,h} \quad (= \mathrm{EAVHD}_i/365). \tag{10}$$

Daily delay is left blank if the data has no weekend days. Both use AADT, an all-days average; weekday traffic is usually somewhat higher, so the weekday figure is, if anything, slightly low.

## 8 Rankings and hotspots

**In plain terms**

We rank road sections by delay per mile, so long sections do not win just by being long, and separately find the single worst mile on each road. A section must be at least a mile long with at least four real segments to be ranked.

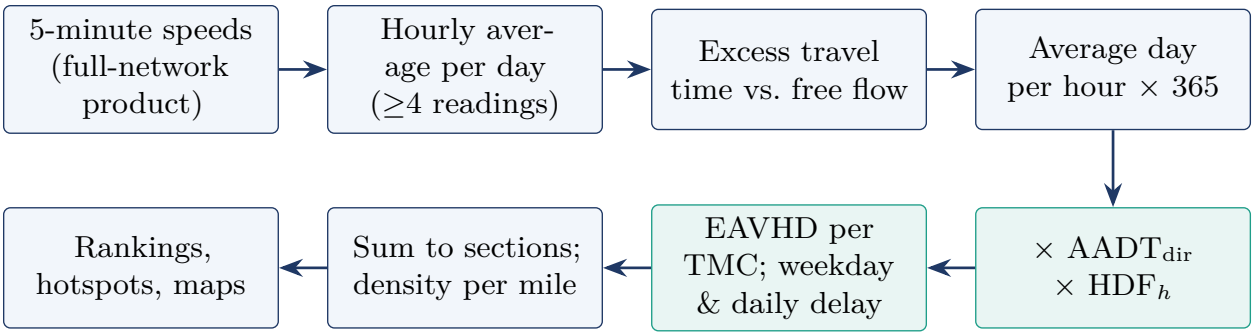



**Figure 13:** From speeds to delay. Blue steps use measured speeds; green steps bring in estimated traffic.

## 8.1 Section delay and density

For a section $c$ (both directions):

$$\text{EAVHD}_c = \sum_{i \in c} \text{EAVHD}_i, \qquad \text{Density}_c = \frac{\text{EAVHD}_c}{L_c}, \tag{11}$$

where $L_c$ is the length of the longer analysed direction (the road's length, not both directions added). A section is eligible for the ranking if it is at least 1.0 mile long and has at least 4 TMCs of 0.05 mile or more (shorter "stubs" do not count). The share of a section's delay in its single largest TMC is also checked; above 60% the section is flagged for review (not excluded).

## 8.2 Hotspots

A one-mile window slides along each road; the window with the most delay is that road's hotspot. A half-mile window is computed as a sensitivity check. Hotspots answer a different question from sections: "where is the delay concentrated?" rather than "which stretch performs worst on average?".

**Table 7:** Different measures answer different questions.

| Measure | Favours |
|---|---|
| Annual delay per mile (section rank) | Consistently congested sections |
| Total annual delay | Long, busy sections |
| Worst one-mile window (hotspot) | Concentrated bottlenecks |
| Recurrence | Congestion that happens on most days |
| Average weekday / daily delay | Plain-unit comparison of daily burden |

## 8.3 Regional delay in 2025

An average weekday carries about 10% more delay than the average day and about 47% more than an average weekend day. Because all three use the same all-days AADT, the weekday figure is probably slightly low and the weekend figure slightly high; the true weekday–weekend gap is likely larger.

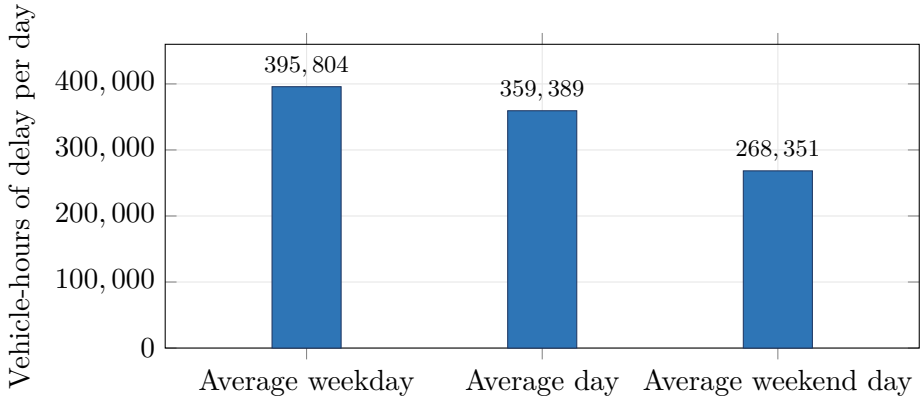



**Figure 14:** Estimated arterial delay per day. The weekend figure is derived from the other two: $(7 \times 359{,}389 - 5 \times 395{,}804)/2$. Annual total: 131,177,049 vehicle-hours (= average day × 365).

### 8.4 How stable is the ranking?

The ranking was recomputed with two very different hourly traffic profiles: a flat profile (equal traffic every hour) and a synthetic commuter profile with sharp peaks. In the run before the network expansion, both kept the ranking almost unchanged:

| Alternative profile | Rank correlation (Spearman) | Top-20 overlap |
|---|---|---|
| Flat (uniform) | 0.999 | 20 / 20 |
| Synthetic commuter peaks | 0.999 | 20 / 20 |

## 9 Congestion patterns (COPs)

**In plain terms**

Some roads are slow all day, some only in the evening rush, some hardly ever. We describe each segment by the shape of its typical day and let the computer group similar shapes into four patterns. This tells us not just how much delay a road has, but what kind of problem it is.

### 9.1 What describes a segment's day

Each TMC is described by 19 numbers, all relative to its own free-flow speed (so a 35 mph and a 55 mph road with the same shape look alike):

- its typical speed ratio in each hour from 6 AM to 8 PM (15 values), capped at 1.2 so odd baselines cannot form their own group (the cap applies only here, not to delay);
- its lowest ratio in the morning (6–10 AM) and in the afternoon (3–7 PM);
- the number of daytime hours below 0.70;
- its recurrence.

The 19 values are standardized (each scaled to the same spread) before grouping. No smoothing is applied in production; a three-hour smoothing was tested only in a separate experiment.

## 9.2 How the groups are formed: hierarchical clustering

Hierarchical clustering starts with every segment on its own and repeatedly joins the two groups whose daily shapes are most alike, until four groups remain, like building a family tree from individuals up. The implementation is scikit-learn's `AgglomerativeClustering` with its default settings: Euclidean distance and Ward linkage, which at each step joins the pair that adds the least variation within groups. K-means (four centres, each segment assigned to the nearest, repeated until stable) is run on the same data, and the method with the better separation is kept.

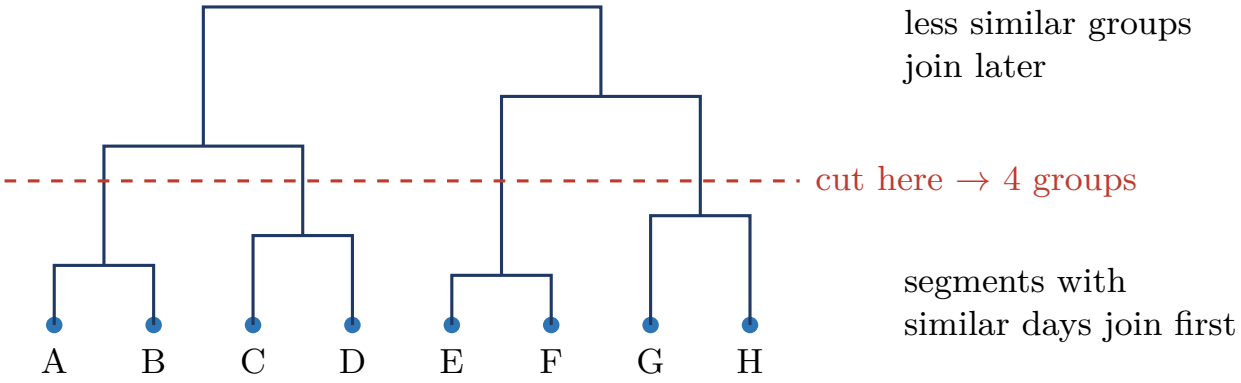



**Figure 15:** Schematic of hierarchical clustering (illustrative, eight segments). The tree is cut where four groups remain.

## 9.3 How the four groups get their names

The names come from fixed rules applied to each group's average day, not from judgment:

| Pattern | Rule on the group's average day |
|---|---|
| COP-1 Sustained All-Day Congestion | at least 65% of daytime hours below 0.70 |
| COP-4 Mostly Uncongested | at most 5% of daytime hours below 0.70, and both AM and PM minima at least 0.85 |
| COP-2 PM-Dominant Congestion | PM minimum below 0.70 and at least 0.05 lower than the AM minimum |
| COP-3 Moderate Daytime Slowdown | any other pattern |

## 9.4 How well the groups separate

The **silhouette** score compares, for each segment, how close it is to its own group with how close it is to the next-nearest group, from −1 (wrong group) through 0 (overlapping) to +1 (perfectly

separate):

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \qquad \bar{s} = \frac{1}{n}\sum_i s(i), \tag{12}$$

where $a(i)$ is the average distance to the segment's own group and $b(i)$ to the nearest other group.

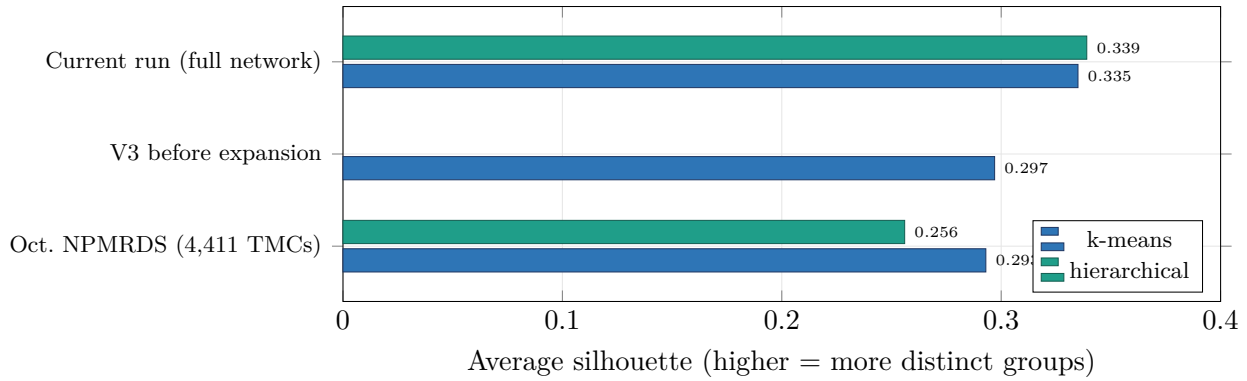



**Figure 16:** Separation of the four patterns across runs (four groups in each). The runs differ in data, network and features, so this is not a like-for-like contest between methods; within the current run the two methods are nearly tied and hierarchical was kept. (The earlier V3 run reported only its chosen k-means score.)

A value of 0.34 means the patterns are **moderately** distinct, typical for traffic patterns that shade into each other. It does not prove the patterns have different causes.

## 9.5 Results: the four patterns in 2025

**Table 8:** Congestion patterns, current run.

| Pattern | TMCs | % of TMCs | Miles | % of miles |
|---|---|---|---|---|
| COP-1 Sustained All-Day Congestion | 519 | 6.0% | 126 | 2.0% |
| COP-2 PM-Dominant Congestion | 1,808 | 20.8% | 739 | 11.5% |
| COP-3 Moderate Daytime Slowdown | 3,345 | 38.5% | 2,338 | 36.4% |
| COP-4 Mostly Uncongested | 3,015 | 34.7% | 3,212 | 50.1% |
| **Total** | **8,687** | | **6,415** | |

**What the patterns show.**

- **Half of the arterial mileage is mostly uncongested** (COP-4, 3,212 miles), largely the newly added suburban and rural roads.
- **Sustained all-day congestion is short and concentrated:** 519 segments but only 126 miles, an average of about 0.24 mile per segment, typical of closely spaced signals in dense urban areas.
- **The evening peak dominates where congestion is recurrent:** 739 miles are PM-dominant (COP-2).
- **The largest group by segments is a moderate daytime slowdown** (COP-3, 2,338 miles): reduced speeds through the day without the sustained congestion of COP-1.

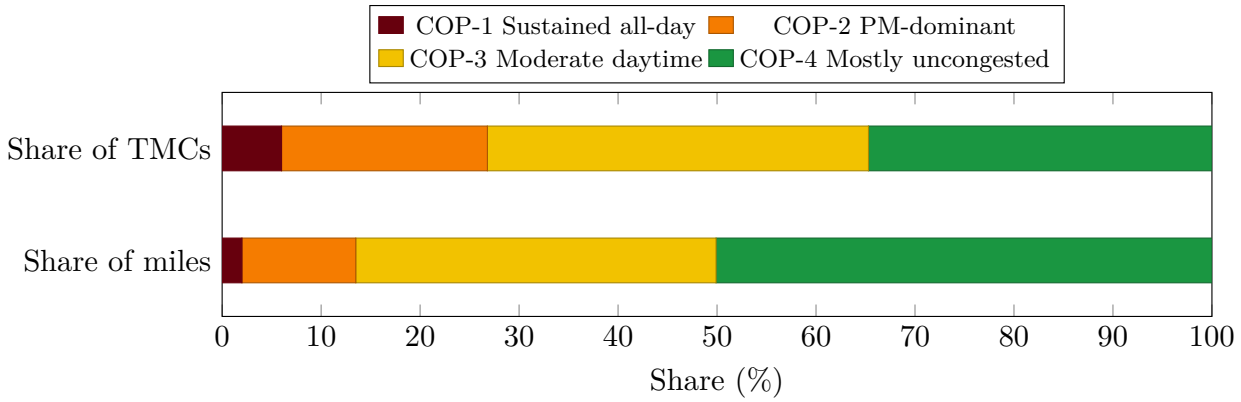



**Figure 17:** The four patterns by number of segments and by mileage.

Each pattern's share of TMCs and of miles differs because segment lengths differ: congested urban segments are short, uncongested rural ones long.

**To be filled from the production run**

Each pattern's average day (mean speed ratio for each hour, 6 AM–8 PM) from the centroid table in the V3 run log, to add a line chart of the four average days.

## 10 Area type and severity: exploratory tests

A separate script (`cbi_area_type_severity_tests.py`) asks whether congestion severity differs by area type. It is exploratory and does not feed the delay or the rankings.

- **Spearman and Kendall rank correlations**: does severity tend to rise or fall as areas become more urban? Because area-type codes are categories, the order used must be stated.
- **Kruskal–Wallis test**: do severity values differ between area-type groups at all? A follow-up comparison of each pair of groups uses the Holm correction for testing several pairs.
- **Partial correlations and robust regression**: does any difference remain after allowing for traffic volume and road class?

Segments on the same road are not independent, so p-values here are approximate. No area-type effect is reported as a finding in this edition.

## 11 Freeway–arterial coupling

**In plain terms**

For each arterial hotspot near a freeway, we test whether freeway congestion happens at the same time as, or an hour or two before, congestion on the arterial more often than chance would predict. This shows association, not cause.

### 11.1 Method

1. Each hotspot is matched to a freeway: by the route named in its interchange (e.g. "I-285" in "I-285/GA-407/Lake Hearn Dr"), otherwise by proximity. Only about one mile of freeway around the matched point is used.
2. For every day and hour, each side is marked congested or not (below 70% of its own free flow).
3. The two sides are compared at lags of 0, 1 and 2 hours (freeway first). Tests with fewer than 20 paired hours are not run.

The strength of association is the **phi coefficient** of the two-by-two table of counts $a, b, c, d$:

$$\phi = \frac{ad - bc}{\sqrt{(a+b)(c+d)(a+c)(b+d)}}, \tag{13}$$

read as negligible (below 0.10), weak (0.10–0.30), moderate (0.30–0.50) or strong (0.50 and above). Significance uses **Fisher's exact test**; because about 60 tests run at once (20 hotspots × 3 lags), p-values are adjusted with the **Benjamini–Hochberg** false-discovery-rate method and a pair counts as significant when $q < 0.05$:

$$\text{largest } k \text{ with } p_{(k)} \le \frac{k}{m}\,\alpha, \qquad \alpha = 0.05. \tag{14}$$

### 11.2 Results and their limits

In the run before the expansion, 20 hotspots were matched to a freeway and **19** showed significant coupling (60 tests). Examples: Pleasant Hill Rd with I-85 southbound at a one-hour lag ($\phi \approx 0.41$–0.44, moderate), and GA-138 with I-20 westbound ($\phi \approx 0.04$, significant but negligible).

**Read with care**

With thousands of paired hours, even negligible associations become "significant". Read phi, not only $q$. Hourly observations are also correlated with each other (rush hour on both roads), which makes p-values optimistic. Proposed safeguards: count only $\phi \ge 0.10$, limit proximity matches to 1 mile, add day-level permutation tests and analyse weekdays and weekends separately.

**To be filled from the production run**

Current-run coupling: number of hotspots matched, number significant, and the phi and lag of each (V3 log and `arterial_freeway_coupling`).

## 12 The Arterial Speed Explorer

The interactive Explorer shows the hourly results on nine linked views. It supports review and communication; agreement between views is a consistency check, not proof of field accuracy.




Fixed speed classes (share of free-flow speed), identical every run and year.

| View | Shows |
|---|---|
| 1 Network | Every analysed arterial and its section |
| 2 Speed Classes | Miles in each speed class, hour by hour |
| 3 Time of Day | Map of speed classes for any hour, weekday or weekend |
| 4 Delay & Volume | Delay and traffic by segment and hour |
| 5 Direction | The two directions of a road compared |
| 6 Time–Space | Speed along a section through the day; header shows annual delay per mile, average weekday and daily delay |
| 7 Segment Detail | One TMC: hourly speeds, readings, annual delay, average weekday and daily delay |
| 8 3D Delay Walls | Delay as wall height along the roads |
| 9 Hours Compared | Two hours side by side on one scale |

# 13 Data engineering and computing

**In plain terms**

Two billion speed readings must be loaded, checked and read many times. This section records how that was done safely and quickly. None of these steps changes what the analysis computes, only how fast and how reliably it runs.

## 13.1 Loading the full-network data

1. **Staging.** Both CSV files were loaded into a temporary (unlogged) table. The fastest method split each file into 16 parts and loaded them with 16 server-side `COPY` commands at once (Figure 18).
2. **Permanent table (migration 022).** `probe_readings_arterial_full` is crash-safe (logged) and keeps six columns. Readings present in both downloads were kept once (higher confidence). Each month was loaded in one transaction together with an entry in a load log, so a month is either fully loaded or not at all, and a rerun skips finished months. Each month took 12–16 minutes.
3. **Checks.** Total and monthly counts, TMC count, date range and arterial coverage were verified, the table was indexed on (TMC, time) and its statistics refreshed, and the staging table was then removed.

## 13.2 Running the analysis in parallel

The workstation has 24 cores (32 threads) and 128 GB of memory. Three changes let the database use them:

After the changes, the plan for the heaviest queries is a parallel scan with nine workers (a `Gather Merge` over a `Parallel Seq Scan` and `Parallel Hash Join`), confirmed while the analysis ran.

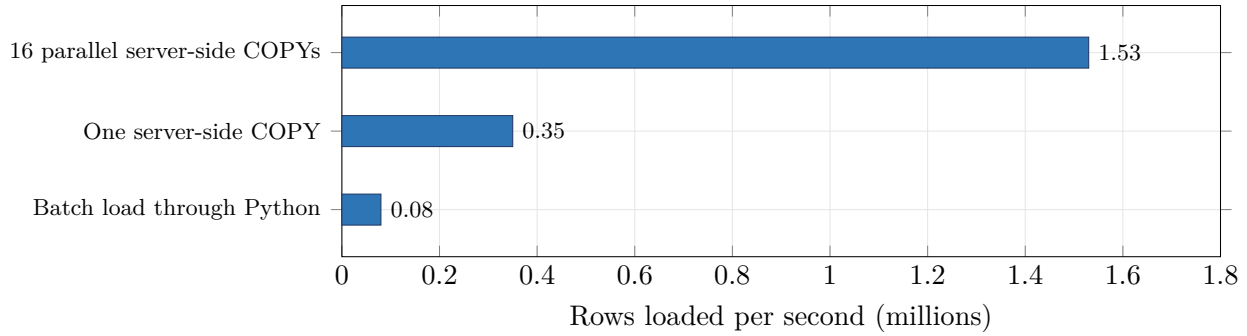



**Figure 18:** Loading speed. One `COPY` is limited to one processor core (about 18 MB/s); 16 in parallel loaded the 19-county file (1.35 billion rows) in 14.7 minutes.

| Change | Why |
|---|---|
| Database settings sized to the machine | Up to 32 worker processes, 24 parallel workers (12 per query step), 16 GB cache, 1 GB working memory per sort step |
| Queries sent as plain `SELECT`s | The data library had wrapped every query in `COPY`, which never runs in parallel |
| Arterial list stored as an analysed table (`v3_meta`) | Computed inline, the list was estimated at about 200 segments instead of about 8,700, so the database looked them up one at a time |

## 14 Checks, validation and what remains

**In plain terms**

We have checked that the data loaded completely, that the two speed products line up in time, that the road-class rules find known arterials, and that thin data no longer creates false congestion. We have *not* yet checked the estimated traffic against independent counts, or the top locations in the field. Those are the next steps.

## 14.1 What has been checked

| Check | Evidence | What it shows |
|---|---|---|
| Data completeness | Monthly and total counts, TMC count, date range, duplicates removed | The data loaded fully |
| Product comparison | Time-shift test; speed and reference differences | Timestamps align; products differ |
| Classification | 97% of known arterial miles matched; volume floor benchmark | The arterial rules are plausible |
| Sparse-hour rule | Mableton example; 130 TMCs on NPMRDS | False night congestion removed |
| Ranking stability | Spearman 0.999, top-20 overlap 20/20 under two profiles (pre-expansion run) | Order is robust to the hourly profile |
| Pattern separation | Silhouette 0.339 vs. 0.335 | Groups are moderately distinct |
| Coupling | Fisher tests with FDR control | Association, not cause |
| Explorer | Nine linked views | Internal consistency |

## 14.2 Three kinds of evidence

- **Calibration** means choosing a setting against a benchmark. The 0.70 threshold, 1-mile section minimum, 1,500 volume floor, 20 m matching tolerance and 4-readings rule are reasoned choices, some tested against benchmarks (the volume floor, the matching tolerance), but none was optimized against field data.
- **Verification** means the software does what the method says: profiles sum to one, units are right, sections reproduce, Explorer figures match the database.
- **Validation** means comparing results with data *not* used to build them. This has not yet been done region-wide.

## 14.3 Recommended validation programme

| Question | Test | Use |
|---|---|---|
| How accurate is hourly traffic? | Compare Replica × HDF with count stations not used for the HDF; report bias, mean error and percentage error by area type, road class and hour | Correct systematic over- or under-estimates |
| How much does the volume source matter? | Recompute delay with Replica vs. the fallback where both exist; compare section ranks | Separate speed-driven from volume-driven results |
| Does one regional HDF fit all roads? | Leave-one-station-out tests; weekday vs. weekend; urban vs. suburban profiles | Decide on separate profiles |
| Are free-flow speeds credible? | Compare 4 AM free flow with posted speeds and independent travel-time runs | Detect over- or under-stated delay |
| Are hotspots real? | Compare top hotspots with queue observations, cameras, signal data or field visits | Field confirmation |
| Are ranks robust? | Alternative windows, profiles, sources and annualization on common TMCs; rank correlation and top-$k$ overlap | Identify stable priorities |
| Are patterns stable? | Vary the number of groups, linkage and samples; compare assignments | Confirm the four patterns |
| Is coupling real? | Day-level permutation tests; separate weekday/weekend; minimum effect size | Avoid spurious significance |

### 14.4 Limitations and open decisions

- The full-network product fills and smooths sparse intervals and may soften brief, sharp queues.
- Hourly traffic is modeled (Replica AADT × a regional profile), not counted; errors in it scale delay directly.
- The 1,500 volume floor may admit some collectors and drop some very quiet rural arterials. A stricter 2,500 floor for rural state routes on OSM secondary roads is a sensitivity option.
- About 170 miles of officially local-class roads behave like arterials and await review; 1,246 TMCs lack a HERE FRC.
- Weekday delay uses all-days AADT, so it is probably slightly low.
- Rankings from different phases are not comparable without restricting to common segments.

## 15 Conclusion

The arterial CBI analysis has become a complete, consistent, year-round picture of arterial congestion in the region: nearly every arterial is now covered by one speed source; the road list was completed with three independent checks; thin data no longer creates false congestion; delay is annualized fairly and reported in plain daily units; and area types use the correct ABM definition. The four congestion patterns add the "what kind of problem" to the "how much delay".

For planning and project screening, results are best presented side by side: delay per mile, daily delay, recurrence, congestion pattern and hotspot, each with its data source. The next milestone is not more visualization but evidence: an independent check of the estimated hourly traffic, sensitivity of the rankings to the volume assumptions, and field confirmation of the top locations. Until then

the results are a rigorous tool for comparison and prioritization, not a count of hours lost or proof that freeway congestion causes arterial congestion.

# A Database migrations and scripts

| Item | Purpose |
|---|---|
| 019 / 020 | Class overrides for Peachtree St, Floyd Rd SW, GA-139; Rountree Rd kept out |
| 021 | 2,297 TMCs added by HERE FRC + OSM class + volume |
| 021b | 1,847 connectors kept (explicit class 4); 471 rejected siblings kept out (class 5) |
| 022 | Permanent full-network table `probe_readings_arterial_full`, loaded month by month |
| `cbi_parallel_stage.py` | Splits a large CSV and loads it with parallel server-side `COPY`s |
| `cbi_import_tmc_frc.py` | Loads TMC code, HERE FRC, area type and road shapes locally into the database |
| `cbi_import_tmc_identification.py` | Adds TMCs missing from the metadata (insert only) |
| `arterial_network_coverage.sql` | Network-wide classification audit (summary output) |
| `frc_replica_comparison.sql` | The road-classification comparison |
| `candidate_volume_check.sql` | The candidate traffic-volume comparison |
| `cbi_arterial_intersection_severity.py` | The arterial analysis; settings `PROBE_TABLE`, `ANNUALIZE_TO_FULL_YEAR`, `CX_PROTOCOL` |

# B Summary of Results from PostgreSQL

**Table 9:** Top 20 Arterial Corridor Sections Ranked by Delay per Mile

| Rank | Arterial Corridor | ARC Area Type | Miles | Delay/Mile/Yr (veh-hr) | Avg Weekday Delay (veh-hr/day) |
|---|---|---|---|---|---|
| 1 | PLEASANT HILL RD [AT1 #3] | CBD | 2.04 | 274,216 | 1,526 |
| 2 | Holcomb Bridge Rd (GA-140) [AT3 #14] | Urban Residential | 3.30 | 232,966 | 2,247 |
| 3 | Peachtree Rd (GA-141) [AT1 #1] | CBD | 3.39 | 174,040 | 1,700 |
| 4 | McDonough Rd (GA-20) [AT4 #10] | Suburban Commercial | 2.42 | 170,881 | 1,173 |
| 5 | Barrett Pkwy (GA-5-CONN) [AT2 #1] | Urban Commercial | 4.20 | 169,116 | 1,935 |
| 6 | Piedmont Rd [AT1 #4] | CBD | 6.59 | 138,104 | 2,703 |
| 7 | Peachtree St (US-19/GA-9) [AT1 #2] | CBD | 2.87 | 135,127 | 1,180 |
| 8 | Scenic Hwy (GA-124) [AT4 #4] | Suburban Commercial | 8.60 | 133,217 | 3,347 |
| 9 | Jesse Jewell Pkwy (GA-369) [AT2 #2] | Urban Commercial | 2.43 | 126,127 | 962 |
| 10 | N DRUID HILLS RD [AT2 #3] | Urban Commercial | 3.19 | 121,743 | 1,219 |
| 11 | MOUNTAIN INDUSTRIAL BLVD [AT5 #2] | Suburban Residential | 2.13 | 119,081 | 789 |
| 12 | Stone Mountain Hwy (US-78) [AT5 #14] | Suburban Residential | 3.23 | 117,809 | 1,096 |
| 13 | PLEASANT HILL RD [AT4 #4] | Suburban Commercial | 6.84 | 116,015 | 2,312 |
| 14 | Buford Dr (GA-20) [AT5 #20] | Suburban Residential | 14.80 | 111,313 | 4,585 |
| 15 | PLEASANT HILL RD [AT2 #2] | Urban Commercial | 2.87 | 109,924 | 924 |
| 16 | Tara Blvd (GA-54) [AT4 #7] | Suburban Commercial | 3.10 | 108,677 | 964 |
| 17 | Buford Dr (US-23) [AT4 #15] | Suburban Commercial | 2.74 | 108,162 | 873 |
| 18 | Northside Dr (US-19) [AT2 #7] | Urban Commercial | 3.51 | 106,665 | 1,109 |
| 19 | Moreland Ave (US-23) [AT3 #16] | Urban Residential | 6.08 | 106,376 | 1,807 |
| 20 | University Pkwy (GA-316) [AT5 #1] | Suburban Residential | 6.43 | 105,956 | 2,284 |

**Table 10:** Hierarchical Clustering Results: Arterial Congestion Patterns

| Congestion Pattern | TMCs | Miles |
|---|---|---|
| COP-1 - Sustained All-Day Congestion | 519 | 126 |
| COP-2 - PM-Dominant Congestion | 1,808 | 739 |
| COP-3 - Moderate Daytime Slowdown | 3,345 | 2,338 |
| COP-4 - Mostly Uncongested | 3,015 | 3,212 |

**Table 11:** Regional Arterial Vehicle-Hours of Delay

| Average Weekday (veh-hr/day) | Average Daily (veh-hr/day) | Annual (veh-hr/year) |
|---|---|---|
| 395,804 | 359,389 | 131,177,049 |

**Table 12:** Directional AADT Source Assignment and Coverage

| Directional AADT Source | TMCs | Miles |
|---|---|---|
| Replica 2025 directional | 4,125 | 3,793 |
| Replica 2025 (OSM secondary, 021 match) | 1,953 | 2,030 |
| Replica 2025 (OSM primary, 021 match) | 333 | 349 |
| Replica 2025 (OSM tertiary, full file) | 210 | 93 |
| Replica 2025 (adjacent 021 TMC, 021b) | 1,846 | 66 |
| NPMRDS aadt_unidir (Replica >2x off, review) | 148 | 50 |
| Replica 2025 (OSM motorway, full file) | 33 | 18 |
| Replica 2025 (OSM trunk, 021 match) | 11 | 8 |
| Replica 2025 (OSM motorway_link, full file) | 11 | 6 |
| Replica 2025 (OSM residential, full file) | 10 | 1 |
| Replica 2025 (OSM trunk_link, full file) | 2 | 1 |
| NPMRDS aadt_unidir (no Replica match) | 4 | 0 |
| Replica 2025 (OSM unclassified, full file) | 1 | 0 |

# C Sources

1. ARC CBI Comprehensive Technical Methodology, October 2026.
2. Arterial Network Expansion and Full-Network Probe Data: Technical Note, November 2026 (controls where documents differ).
3. The V3 arterial analysis code (settings and formulas in this report were checked against it).
4. ARC Activity-Based Model Data Dictionary, February 2022 (area-type definitions).
5. Run logs and database checks recorded during the November 2026 work.